

VA-ATACOM Supplementary Material: The DT-margin Heuristic Baseline

Anonymous submission

Table 1: DT-margin ablation on Goal (5 seeds).

Config	α	h	Mean $\bar{C}$	GO/5
Velocity-adaptive only	0.3	0	2.90	5/5
BRT-only	0.0	3	2.35	4/5
DT-margin (full)	0.3	3	0.96	5/5

The DT-margin Heuristic Baseline

VA-ATACOM’s velocity augmentation has a heuristic predecessor we call the **DT-margin** filter — the “DT-margin (heuristic)” row of Table 1 in the main paper. It inflates the keepout radius linearly with speed, $r_{\text{eff}}(\mathbf{v}) = r_{\text{base}}(1 + \alpha \|\mathbf{v}\|)$, projects out the inward radial component, and adds the same sim-BRT reward shaping. As shown in the main paper’s design study, this linear margin is the *first-order* approximation of VA-ATACOM’s quadratic braking constraint; under fair hazard-radius matching it reaches 13/15 GO — two Push seeds short of VA-ATACOM’s 15/15, illustrating that the linearisation almost captures the right behaviour but leaves a residual gap that only the exact quadratic constraint closes. We characterise it here for completeness — its two tunable knobs (α and the BRT horizon h) and their interaction with the control rate. These findings motivated, and are superseded by, the parameter-free velocity-augmented manifold of the main paper.

Ablation of the two components

Table 1 (Fig. 1) shows both components contribute: velocity-adaptive margin alone reaches 5/5 GO at higher variance (2.90); BRT alone 4/5 GO (one M1 collapse at $\bar{C}=7.98$); the full heuristic 0.96 cost / 100% GO.

Δt sensitivity

Table 2 corroborates the M1 chord-excursion premise of the main paper’s failure-mode analysis (and hence Prop. 4): continuous-time ATACOM achieves zero cost at $\Delta t \leq 0.02\text{s}$ but fails at $\Delta t \geq 0.05\text{s}$, the phase transition matching the M1 threshold $\Delta t \|u\| \gtrsim \sqrt{d_{\text{safe}}} \cdot r$. The DT-margin heuristic tracks the safe regime at $\Delta t \in \{0.10, 0.20\}\text{s}$ with $h=3$. At $\Delta t=0.05\text{s}$ the picture is nuanced: the horizon rule prescribes $h^*=4$, but a 5-seed sweep finds $h=3$ at 3/5 GO (bimodal), $h=4$ at 0/5, and $h=5$ at 3/5 (one seed at 37.05). We attribute

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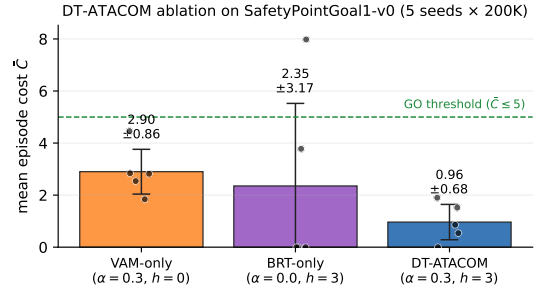


Figure 1: DT-margin ablation: both components are necessary. Mean episode cost $\bar{C}$ on SafetyPointGoal1-v0 (5 seeds $\times$ 200K env steps), bars are ± 1 std. VAM-only ($\alpha=0.3, h=0$) addresses M1 chord excursion only; BRT-only ($\alpha=0, h=3$) anticipates M2 drift but one seed collapses via M1; DT-margin ($\alpha=0.3, h=3$) combines both for lowest mean (0.96) and 5/5 GO.

Table 2: DT-margin cost vs. time step ($\bar{C} \pm \sigma$, 5 seeds, 50K steps).

Δt	ATACOM	DT-margin ($h=3$)	Note
0.01s	0.00	3.02	ATACOM OK
0.02s	0.00	0.00	ATACOM OK
0.05s	39.95	4.96 $\pm$ 6.08 (3/5 GO)	Basin regime
0.10s	13.68	2.60	DT-margin OK
0.20s	—	1.33 $\pm$ 1.50	$h \gg h^*$

this to a *BRT-shaping basin*: near the penetration horizon the lookahead reports unavoidable penetration along multiple directions every step, integrating spurious penalty into the policy gradient and trapping training.

α/h sensitivity

Sweeping α and h on Goal (5 seeds $\times$ 200K):

Only configurations at least one step above the minimum on *both* axes ($\alpha \geq 0.1, h \geq 2$) reach 5/5 GO; setting $\alpha=0$ drops to 4/5 with a seed at $\bar{C}=8.46$. The over-provisioned config ($\alpha=0.5, h=5$) *regresses* to 4/5 ($\bar{C}=9.26$ outlier) — large- h BRT shaping can integrate spurious penalty (see the Δt sensitivity above) and destabilise training. The DT-margin recipe

Table 3: DT-margin α/h sensitivity on Goal (5 seeds $\times$ 200K).

Config	α	h	mean $\bar{C}$	GO/5
no margin	0.0	2	3.37 ± 2.81	4/5
short horizon	0.05	1	1.94 ± 1.78	4/5
both at minimum	0.05	2	2.36 ± 2.60	4/5
above minimum	0.1	2	1.30 ± 1.66	5/5
α generous	0.3	1	1.72 ± 1.46	5/5
paper choice	0.3	3	0.96 ± 0.68	5/5
over-provisioned	0.5	5	3.00 ± 3.38	4/5

is thus robust only with margin on both knobs; VA-ATACOM eliminates α entirely, replacing it with the calibrated $a_{\max}$.

Statistical Significance

Safety-Gymnasium (Table 1 of the main paper). We pair every (task, seed) of each baseline against the same (task, seed) of VA-ATACOM (15 paired episode-cost samples per baseline; 11 baselines $\times$ 15 = 165 paired tests in total). One-sided Wilcoxon signed-rank tests (alternative: baseline cost > VA-ATACOM cost) give:

Baseline	p -value	sig.
PPO baseline	3.1×10^{-5}	††
ATACOM (null)	3.1×10^{-5}	††
ATACOM-VD	3.1×10^{-5}	††
ATACOM-S	3.1×10^{-5}	††
ATACOM-LA	3.1×10^{-5}	††
HOCBF	3.1×10^{-5}	††
DCM	3.1×10^{-5}	††
Predictive ATACOM	4.9×10^{-4}	††
CBF-QP	3.1×10^{-5}	††
PPO-Lag	3.1×10^{-5}	††
DT-margin (1st-order approx.)	2.9×10^{-3}	†

†: $p < 0.05$, ††: $p < 0.001$

All 11 baselines are significantly worse than VA-ATACOM ($p < 0.05$); 10 are extremely significant ($p < 0.001$). DT-margin, the first-order Taylor approximation of VA-ATACOM, has the smallest gap ($p=0.003$) — consistent with the main paper’s claim that the linearisation captures most of the safe behaviour but leaves a residual gap that only the exact quadratic constraint closes (10/15 paired cells where DT-margin > VA-ATACOM, 4/15 ties at 0). A Friedman omnibus test across all 12 methods rejects equality with $\chi^2=97.05$, $p=6.8 \times 10^{-16}$ (df=11).

Webots transfer (120 paired trials). For the Webots E-puck study (§ Sim-to-Real of main paper) we apply the McNemar test for paired binary outcomes (deep penetration ≥ 1 cm yes/no) on the same (start, goal, world, controller) configuration with the filter on vs. off:

The filter *saves* 23 paired trials (filter on safe, filter off deep) and *hurts* 0 paired trials (filter on deep, filter off safe). The number-needed-to-filter is $N/\text{saves} = 120/23 \approx 5.2$: on

Setting	N	on-deep	off-deep	saves	p
P-controller (3 worlds)	60	1	14	13	1.2×10^{-4}
PPO transfer (3 worlds)	60	0	10	10	9.8×10^{-4}
All 120 trials	120	1	24	23	1.2×10^{-7}

average one in every five trajectories needs the filter to avoid deep penetration. The one VA-ATACOM deep case (dense, P-controller, triple-obstacle pocket) is *also* deep without the filter, so the filter never makes things worse.

Generalisation Beyond Point-Robot Dynamics

To probe VA-ATACOM’s scope across different mobile-robot dynamics, we evaluate the unchanged filter on `SafetyCarGoal1-v0`, which replaces the Point agent’s holonomic velocity actions with a bicycle-like throttle+steering. All other env parameters (8 hazards of radius 0.2, BRT frontend, 5 seeds, 200K steps) match the main Table 1.

Method	mean $\bar{C}$	GO/5
PPO (no filter)	23.20	1/5
ATACOM (null-space)	9.20	1/5
DT-margin (heuristic)	13.02	1/5
VA-ATACOM	23.92	1/5

Every method is NO-GO under the 200K budget, but the relative ordering *inverts* compared with the Point-robot Table 1: on Car, the null-space ATACOM (9.20) and DT-margin (13.02) beat VA-ATACOM (23.92).

The cause is structural and identified in our main paper: VA-ATACOM’s tangent-cone projection is currently implemented with the `action_form='diff_drive'` constraint (§ VA-ATACOM Algorithm), which assumes *forward speed and angular rate are independent actuators*. Car’s bicycle model couples them through the wheelbase, so reducing forward command also reduces steerability — the filter’s safety brake removes the very degree of freedom the agent needs to turn away from the obstacle. The null-space ATACOM projection, by contrast, projects onto the Jacobian null-space of the position constraint, which is independent of the actuation parameterisation and degrades gracefully on Car.

This is a clear scope clarification for VA-ATACOM, not a refutation of Prop. 4 itself: the manifold and its discrete-time invariance theorem hold; the bicycle-Car case requires a different action-form projection (matching steering Jacobian) for the implementation. Designing a bicycle-aware action form is an obvious follow-up and is listed in the main paper’s Limitations L4.

Computing Infrastructure

All experiments reported in the main paper and this supplementary were run on a single workstation:

- **CPU:** Intel Core i7-13850HX (20 cores, 28 threads).
- **RAM:** 31 GiB DDR5.

- **OS:** Ubuntu 24.04 LTS, Linux kernel 6.17.
- **GPU:** not used — all training and evaluation run *single-threaded on CPU* (OMP_NUM_THREADS=1, MKL_NUM_THREADS=1). Each Safety-Gymnasium 200K-step run takes $\approx 15\text{--}20$ min wall-clock; we parallelise up to 20 such single-threaded runs at once.

Software stack (versions used at experiment time):

- Python 3.10
- PyTorch 2.x (CPU build)
- NumPy 1.23 / matplotlib 3.x
- `safety-gymnasium 1.0` (the Safety-Gym successor; we ran with `SafetyPointGoal1-v0`, `SafetyPointPush1-v0`, `SafetyPointMultiGoal0-v0`)
- `gymnasium 0.28` / `mujoco 2.3`
- Webots R2023b (for the rigid-body validation in main § Sim-to-Real)

Random seeds 0, 1, 2, 3, 4 are used for every (method, task) cell in Table 1 of the main paper. Total compute for the full Table 1 + Webots study is approximately **200 CPU-hours** on the workstation above (~ 10 h wall-clock with 20-way parallelism); the L-shape audit re-run added a further ~ 5 CPU-hours.